

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 8: Applying Hormones: Rooting Cuttings for the School Tree Nursery

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	C	D	C	A	C	D

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share ranking of the four treatments (Where do you think auxin collects in a cutting?)

Q1. In Lesson 7 you learned that auxin is involved in rooting. A hibiscus stem cutting has been cut from the parent plant and has no roots. Where does auxin collect in the cutting so that roots can form?

- A. In the soil around the cutting
- B. At the tips of the leaves
- C. In seeds inside the cutting
- D. At the cut base of the stem**

Correct answer: D. Auxin (IAA) accumulates at the cut base of a stem cutting, and this stimulates root formation. Cuttings do not need seeds to make roots.

USE AFTER: OBSERVE PHASE — Part A: setting up the mini fair test with four labelled containers (identifying the control)

Q2. Madam Achieng's class compares bulb onion extract, Aloe vera gel and cinnamon powder for rooting hibiscus cuttings. They also include a container of plain water with no added substance. Why is plain water included?

- A. It gives the cuttings the hormone they need to root
- B. It makes the test fair by changing the water in all the other containers
- C. It is the control, so we can compare treated cuttings with cuttings that had no added substance**
- D. It washes the cuttings so they stay clean

Correct answer: C. The plain water cuttings are the control. They show how many cuttings root without any added substance, so we can tell whether the treatments made a difference.

USE AFTER: OBSERVE PHASE — Part A: identifying independent, dependent and controlled variables

Q3. In a cuttings fair test, the type of treatment (water, onion, Aloe vera or cinnamon) is changed on purpose, and the number of cuttings that root is measured. Which of these must be a controlled variable (kept the same for all cuttings)?

- A. The percentage rooting after 21 days
- B. The type of treatment used
- C. The number of cuttings that root
- D. The length of each cutting (about 15 cm)**

Correct answer: D. Cutting length must stay the same so it cannot affect rooting. The type of treatment is the independent variable. The number rooted and the percentage rooting are the dependent variable.

USE AFTER: OBSERVE PHASE — Part B: calculating percentage rooting from the 21-day results

Q4. In the class nursery test, 30 hibiscus cuttings were treated with bulb onion extract. After 21 days, 21 of them had rooted. What is the percentage rooting?

- A. 30%
- B. 21%
- C. 70%**
- D. 51%

Correct answer: C. Percentage rooting = (number rooted ÷ total cuttings) × 100 = 21 ÷ 30 × 100 = 70%.

USE AFTER: EXPLAIN PHASE — Explain: CER discussion (Plain water still rooted 9 of 30. What does that tell us?)

Q5. In a nursery test with 30 cuttings per treatment, 9 cuttings in plain water (no added substance) still grew roots. What does this tell us?

- A. The cuttings can make some auxin themselves, so some root even without a treatment**
- B. The test was unfair and the results cannot be used
- C. Cuttings root only when they have seeds
- D. The plain water contained a rooting hormone

Correct answer: A. Plain water has no added hormone, so the cuttings that rooted must have used auxin they made themselves. The treated cuttings rooted more because the substances added to this natural rooting.

USE AFTER: EXPLAIN PHASE — Explain: discussing the limits of the evidence (one trial, one species, no direct measure of auxin)

Q6. Results after 21 days (30 cuttings each): plain water 9 rooted, bulb onion 21, Aloe vera 24, cinnamon 18. A learner says Aloe vera is better than onion. Which statement is the best limit of this evidence?

- A. Rooting can never be linked to auxin because auxin was not measured
- B. Plain water rooted some cuttings, so the whole test is invalid
- C. The difference between 24 and 21 is small, and there was only one trial on one species, so we cannot be fully sure**
- D. Percentages cannot be used to compare treatments

Correct answer: C. Aloe vera (80%) and onion (70%) are close, and the test was done once on one species. We cannot be certain that Aloe is always better. Percentages are a valid way to compare, and the plain water result is normal for a control.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 model with a section on 'Hormones in growth and development'

Q7. Learners add this to the model explaining why Mr. Wafula's maize and beans grew differently, even though they had the same rain and planting date. Which revision is best supported by the cuttings data (plain water 30%, cinnamon 60%, onion 70%, Aloe vera 80% rooting)?

- A. Differences in growth are caused only by rain and planting date
- B. Only seed structure affects how plants grow
- C. Hormones replace the need for water and soil in growth
- D. Differences in growth can be caused by internal factors such as hormones, as well as seed structure, seed condition and environment**

Correct answer: D. The cuttings were given the same conditions but rooted differently depending on the treatment, which shows hormones can influence growth. Hormones work together with seed structure, seed condition and environment.